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CHAPTER 1 • NUMBER • TOPIC 6

Percentages

One idea runs the whole topic: the multiplier. Increase, decrease, reverse, repeat — multiply or divide by it.



$$\frac{3}{10} = 0.3 =$$

30%**6**

VARIANTS

41

PAST-PAPER PARTS

50

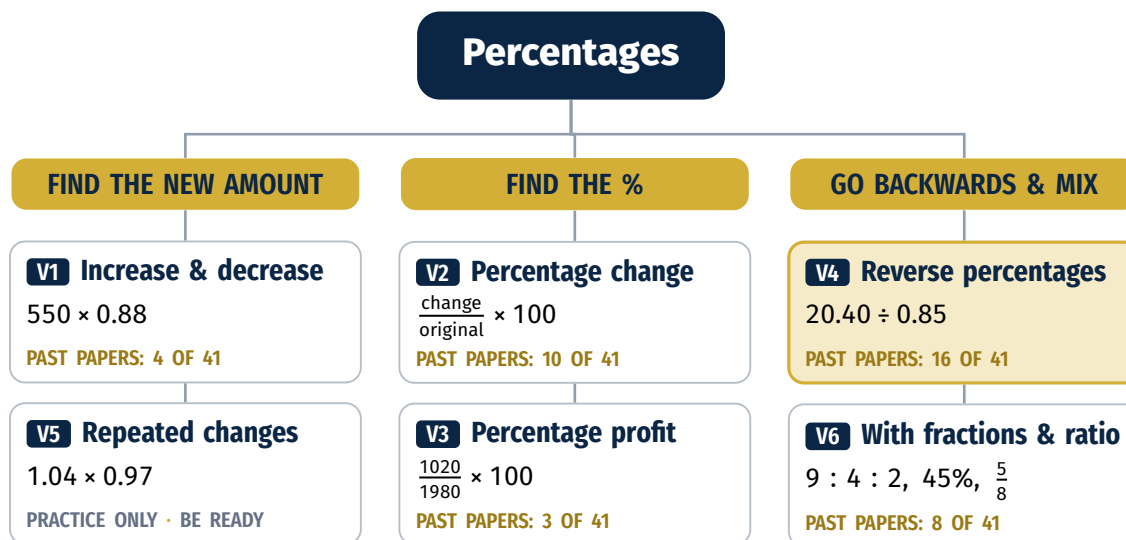
PRACTICE QUESTIONS

YOU WILL MASTER**V1** Increase & decrease**V2** Percentage change**V3** Percentage profit**V4** Reverse percentages**V5** Repeated changes**V6** With fractions & ratio

TOPIC MAP • EVERY VARIANT ON ONE PAGE

Percentages at a glance



Your notes


ONE IDEA

THE MULTIPLIER

+15% → $\times 1.15$ • -15% → $\times 0.85$

 How often does each variant come up?

34 past-paper questions (Jan 2020 → Nov 2025) hold 41 parts; 50 practice parts — grouped by type.

	PAST PAPERS (OUT OF 41)	PRACTICE PARTS
V1 Increase & decrease	4	13
V2 Percentage change	10	5
V3 Percentage profit	3	4
V4 Reverse percentages	16	9
V5 Repeated changes	0	12
V6 With fractions & ratio	8	7

WHAT THIS TELLS YOU Percentages appear on almost every paper — often twice. Reverse percentages (“the sale price is ... find the normal price”) is the single most common question: 16 parts, 3 marks each.

BEFORE YOU START

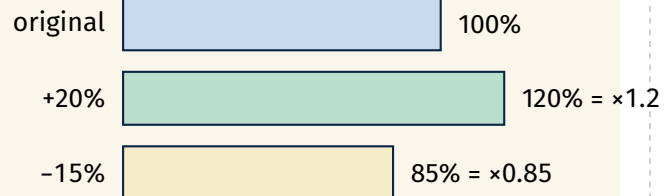
The big idea

**Percent means “out of 100”.**

The original amount is always **100%**.
After a change it is a different percentage of the original — write that as a decimal: the **multiplier**.

+20% → 120% → $\times 1.2$.

-15% → 85% → $\times 0.85$.


 Three questions, three moves
Find the new amount

original $\times$ multiplier

550 down 12%: 550×0.88

Find the original

new $\div$ multiplier

sale 20.40 after -15%:
 $20.40 \div 0.85$

Find the % change

$\frac{\text{change}}{\text{original}} \times 100$

always divide by the
original

 Multipliers to know

+5%

$\times 1.05$

+17.5%

$\times 1.175$

-12%

$\times 0.88$

-2.5%

$\times 0.975$

**CALCULATOR FIRST, THEN CHECK**

Every percentage question is one calculator line: $550 \times 0.88 =$. Check “reverse” answers by going forwards: $24 \times 0.85 = 20.40 \checkmark$.

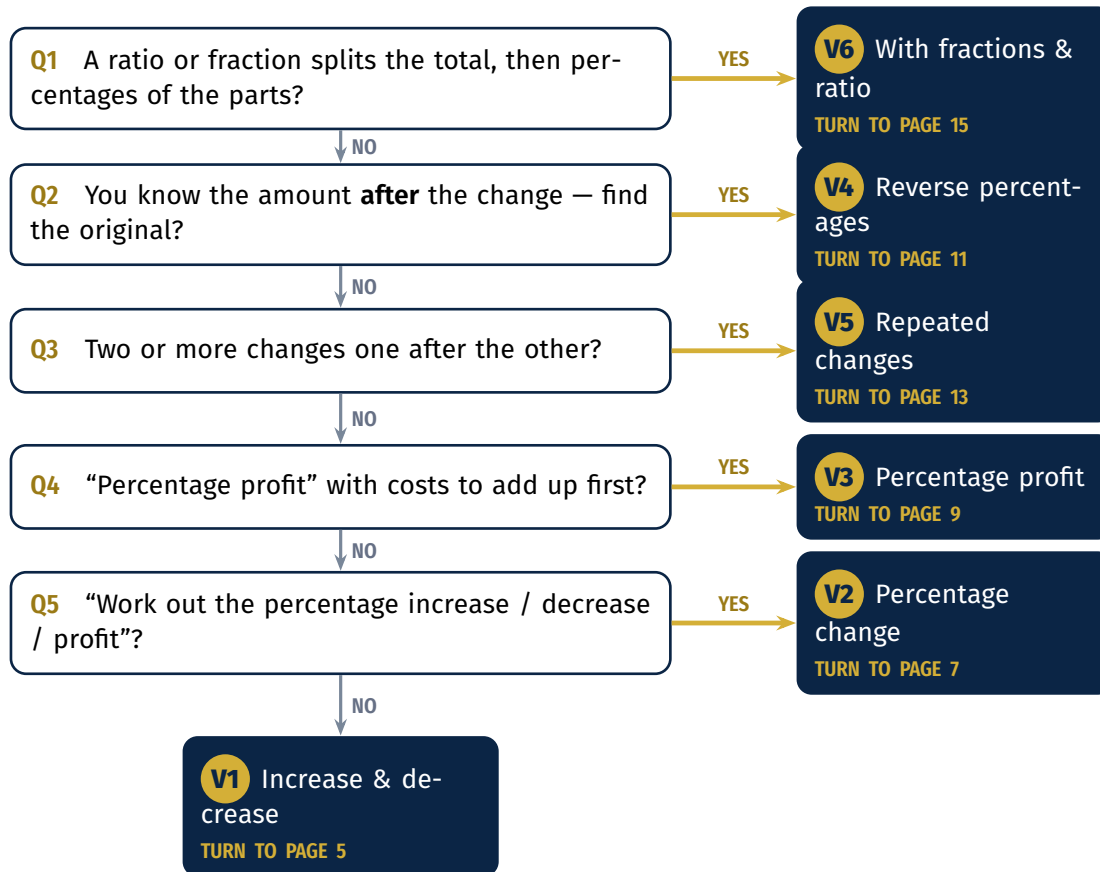
COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?



Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant – and the page to turn to.

**COLOUR CODE IN THE WORKING**

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

ON EVERY VARIANT SPREAD

- | | | | |
|---|-----------------|---|-----------------------|
|  | What is it? |  | When you see → you do |
|  | How do I do it? |  | Watch out |
|  | Worked example |  | Exam tip |
|  | Check yourself |  | Calculator check |

V1

VARIANT 1

Increase & Decrease by a PercentageNew amount = original $\times$ multiplier.

SEEN IN PAST PAPERS

4 / 41

PRACTICE QUESTIONS: 13

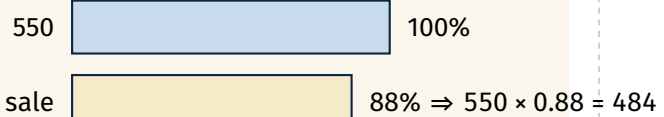


Your notes

Came up in: Jan 2021 P2H Q3(b) • Jan 2022 P2HR Q2(b) • May 2024 P1HR Q3 • Nov 2024 P1H Q7

? What is it?**In plain words**

Increase by 20%: you now have

 $100 + 20 = 120\% \rightarrow \times 1.2$.Decrease by 12%: you have $100 - 12 =$ $88\% \rightarrow \times 0.88$.“Save” or “off” questions: compare the **amounts saved**.**IT LOOKS LIKE**

reduce 550 by 12%

85 + 20% VAT

How do I do it?**1 Percent left**Increase: $100 + p$. Decrease: $100 - p$.**2 Multiplier**Divide by 100: $88\% \rightarrow 0.88$.**3 Multiply**Original $\times$ multiplier.**4 Compare / round**

Money to 2 d.p.; compare options if asked.

ONE MULTIPLIER, ONE STEP

No need to find 12% and subtract.

V1 Increase & Decrease by a Percentage

VARIANT 1 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“decreases all prices by 12%” → $\times 0.88$

“plus VAT at 20%” → $\times 1.2$

“Which coupon should he use?” → **Work out both savings, compare**

WORKED EXAMPLE **PAST PAPER**

January 2022 • Paper 2HR • Q2(b) • 3 marks

Accommodation \$1600, flights \$800. Coupon A: 18% off the accommodation. Coupon B: 12.5% off accommodation **and** flights. Which coupon saves more?

STEP 1 A : $0.18 \times 1600 = 288$ ◀ saving with A

STEP 2 B : $0.125 \times (1600 + 800) =$ ◀ saving with B
 $0.125 \times 2400 = 300$

STEP 4 \$300 > \$288 ◀ compare

ANSWER Coupon B (saves \$300, A saves \$288) ✓ both savings shown

WATCH OUT

✗ B: 0.125×1600

✓ B: 0.125×2400

Read what the percentage is **of**: B covers the flights too.

EXAM TIP

A decision needs both numbers and a sentence: “B, because $\$300 > \288 ”.

Check yourself

calculator first • answers on page 17

1.1

Dr Eslam

A coat costs £85 plus VAT at 20%. Work out the total cost.

Answer:

1.2

Jan 2021 P2H

All prices are decreased by 12%. An armchair was \$550. Work out its sale price.

Answer:

1.3

Dr Eslam

Increase £240 by 15%.

Answer:

V2

VARIANT 2

Percentage ChangeChange $\div$ original $\times 100$.

SEEN IN PAST PAPERS

10 / 41

PRACTICE QUESTIONS: 5



Your notes

Came up in: Jan 2020 P2H Q8(a) • Oct/Nov 2020 P1HR Q5(a) • Jan 2021 P2H Q3(a) • Nov 2021 P2H Q6 • Jan 2022 P2H Q8(a) • Jan 2022 P2HR Q2(a) • May 2023 P1H Q3 • Jun 2023 P2HR Q6(a) • Nov 2023 P1H Q3 • Nov 2024 P2H Q4(b)

? What is it?**In plain words**

How big is the change
compared with where you started?

$\frac{\text{change}}{\text{original}} \times 100$. Profit is a change too:

$\frac{\text{profit}}{\text{cost}} \times 100$.

IT LOOKS LIKE228 314 $\rightarrow$ 231 776\$465 $\rightarrow$ \$520

old 228 314 $\xrightarrow{\text{CHANGE 3462}}$ new 231 776

$$\frac{3462}{228\,314} \times 100 = 1.5\%$$

How do I do it?**1 Original**The value **before** the change (cost price for profit).**2 Change**

New – old (profit = selling – cost).

3 Divide by original $\frac{\text{change}}{\text{original}} \times 100$.**4 Round & say**

To what is asked; say increase or decrease.

DIVIDE BY THE ORIGINAL Never by the new value.

V2 Percentage Change

VARIANT 2 • SEE IT DONE, THEN TRY



Your notes
 WHEN YOU SEE → YOU DO

“Work out the percentage increase” → $(\text{new} - \text{old}) \div \text{old} \times 100$

“percentage profit ... cost \$465, sells \$520” → $55 \div 465 \times 100$

“...more than ...? Is Brigitte correct?” → **Work out both percentages, compare**

 WORKED EXAMPLE **PAST PAPER**

January 2022 • Paper 2H • Q8(a) • 2 marks

The average house price was £228 314 in 2018 and £231 776 in 2019. Work out the percentage increase, correct to 1 decimal place.

STEP 1 original = 228 314 ◀ before the change

STEP 2 $231\,776 - 228\,314 = 3462$ ◀ change

STEP 3 $\frac{3462}{228\,314} \times 100 = 1.516...$ ◀ divide by original

ANSWER 1.5%✓ $228\,314 \times 1.015 \approx 231\,739$ ✓ **WATCH OUT**

$$\times \frac{3462}{231\,776} \times 100$$

$$\checkmark \frac{3462}{228\,314} \times 100$$

The percentage is of the **original** (2018) price.

 EXAM TIP

10 of the 41 past-paper parts. Keep the full calculator value until the last rounding.

 Check yourself

calculator first • answers on page 17

2.1

Jun 2023 P2HR

Pierre earned 400 euros a week. Now he earns 14 euros an hour, 7 hours a day, 5 days. Percentage increase?

Answer:

2.2

Jan 2021 P2H

Gladys buys a table for \$465 and sells it for \$520. Percentage profit (3 s.f.)?

Answer:

2.3

Jan 2022 P2HR

Mary saved \$720 in 2020 and \$78 each month in 2021. The 2021 total was $P\%$ more. Find P .

Answer:

V3

VARIANT 3

**Percentage Profit
(Build the Cost)**

Add up all the costs, add up all the income,
then profit ÷ cost.

SEEN IN PAST PAPERS

3 / 41

PRACTICE QUESTIONS: 4


Your notes

Came up in: Jan 2022 P1H Q4 • Jun 2025 P1H Q2(a) • Jun 2025 P2HR Q4

? What is it?**In plain words**

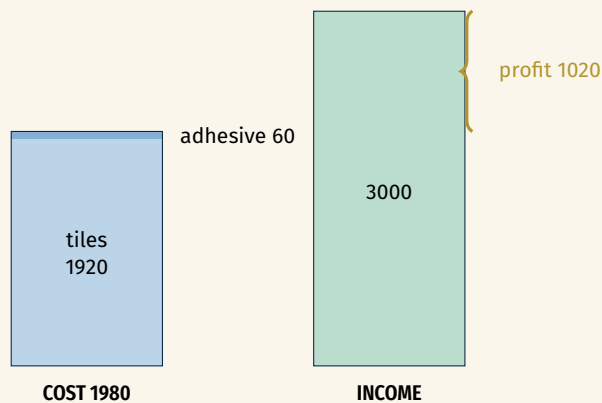
A longer story: the **cost** comes from several parts (boxes, bags, mugs) and so does the **income**. Build each total first.

$$\text{Then \% profit} = \frac{\text{income} - \text{cost}}{\text{cost}} \times 100.$$

IT LOOKS LIKE

tiles + adhesive

small/medium/large mugs

**How do I do it?****1 Total cost**

Every cost: items × price, plus extras.

2 Total income

What is charged or sold.

3 Profit

Income – cost.

4 % profitProfit ÷ **cost** × 100; round as asked.**PROFIT ÷ COST** Cost is the original.

V3 Percentage Profit (Build the Cost)

VARIANT 3 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO“one box for each 1.5 m²”→ **Boxes = area ÷ 1.5 (round up if needed)**

“Work out his percentage profit”

→ **(income – cost) ÷ cost × 100** $\frac{2}{5}$ small, 32% medium, the rest large→ **Find each count first****WORKED EXAMPLE PAST PAPER**

June 2025 • Paper 2HR • Q4 • 5 marks

Floor 45 m². One box of tiles (£64) for each 1.5 m²; 5 bags of adhesive at £12. Joshua charges £3000. Work out his percentage profit (1 d.p.).

STEP 1 $45 \div 1.5 = 30$ boxes : $30 \times 64 = 1920$; $5 \times 12 = 60$ ◀ costs

STEP 1 cost = 1980 ◀ total cost

STEP 3 $3000 - 1980 = 1020$ ◀ profit

STEP 4 $\frac{1020}{1980} \times 100 = 51.51 \dots$ ◀ divide by cost

ANSWER 51.5%✓ $1980 \times 1.515 \approx 3000$ ✓**WATCH OUT**

✗ $\frac{1020}{3000} \times 100 = 34\%$

✓ $\frac{1020}{1980} \times 100$

Percentage profit is on the cost, not on the selling price.

**EXAM TIP**

Write each cost line separately — the method marks are for the totals.

**Check yourself**

calculator first • answers on page 17

3.1

Jun 2025 P1H

Cups cost 6 each to make; 4 per box; 80 boxes sell for 2160. Percentage profit?

Answer:

3.2

Jan 2022 P1H

150 mugs cost £1140. $\frac{2}{5}$ small at £8.50, 32% medium at £11.20, rest large at £14.20. % profit (nearest whole)?

Answer:

3.3

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Bought for £80, sold for £92. Percentage profit?

Answer:

V4**VARIANT 4** MOST COMMON IN THE EXAM

Reverse Percentages

You know the amount after the change:
divide by the multiplier.

SEEN IN PAST PAPERS

16 / 41

PRACTICE QUESTIONS: 9


Your notes

Came up in: Jan 2020 P1HR Q7 • Jan 2020 P2H Q8(b) • Oct/Nov 2020 P1H Q9 • Oct/Nov 2020 P1HR Q5(b) • Jan 2021 P2HR Q6 • Nov 2021 P1H Q8 • Jan 2022 P2H Q8(b) • Jun 2022 P2HR Q8 • Jun 2023 P2HR Q6(b) • Nov 2023 P1H Q6 • May 2024 P1H Q4 • Jun 2024 P2HR Q8 • Nov 2024 P2H Q4(a) • Jun 2025 P1H Q5 • Jun 2025 P2HR Q9 • Nov 2025 P1H Q9

? What is it?

In plain words

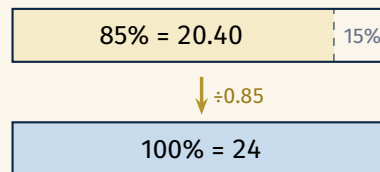
The sale price 20.40 is what is **left after** 15% off, so 20.40 is **85%** of the original.

Go backwards: **divide** by the multiplier. $20.40 \div 0.85 = 24$.

Never take 15% of 20.40 and add it on.

IT LOOKS LIKE

sale price after the increase
the normal price



How do I do it?

1 Which % is known? After -15%: 85%. After +12%: 112%.

2 Multiplier 0.85 or 1.12.

3 Divide Known amount $\div$ multiplier = original.

4 Check forwards Original $\times$ multiplier = the given amount.

BACKWARDS = DIVIDE The most common percentage question.

V4 Reverse Percentages

VARIANT 4 • SEE IT DONE, THEN TRY


 Your notes

WHEN YOU SEE → YOU DO

"The sale price is ... Work out the normal price" → **Sale** $\div (1 - p)$

"After the increase, the cost was ..." → **New** $\div (1 + p)$

"price on the previous Monday" (5% more now) → **Now** $\div 1.05$

WORKED EXAMPLE PAST PAPER

January 2020 • Paper 1HR • Q7 • 3 marks

In a sale, the normal price of a hat is reduced by 15%. The sale price is 20.40 euros. Work out the normal price.

STEP 1 $100\% - 15\% = 85\%$ ◀ which % is known

STEP 2 0.85 ◀ multiplier

STEP 3 $20.40 \div 0.85 = 24$ ◀ divide

ANSWER 24 euros

✓ $24 \times 0.85 = 20.40$ ✓

WATCH OUT

✗ $20.40 \times 1.15 = 23.46$

✓ $20.40 \div 0.85 = 24$

15% of the sale price is not 15% of the original. Divide.

EXAM TIP

16 of the 41 past-paper parts. Always check forwards — it takes five seconds.

Check yourself

calculator first • answers on page 17

4.1 Jun 2022 P2HR

An adult ticket increased by 12% to £18.20. What did it cost before?

Answer:

4.2 Jun 2025 P1H

Prices reduced by 28%. The sale price of a watch is 198 euros. Normal price?

Answer:

4.3 Jun 2024 P2HR

17% off. The sale price of a fridge is 6225 rupees. Normal price?

Answer:

V5

VARIANT 5

Repeated & Combined Changes

Two changes in a row: multiply the multipliers — never add the percentages.

SEEN IN PAST PAPERS

0 / 41

NOT ASKED YET • BE READY
PRACTICE QUESTIONS: 12


Your notes

? What is it?

In plain words

+4% then -3% is **not** +1%. The second change acts on the **new** amount.

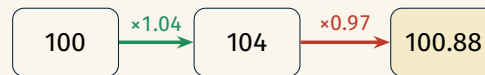
Multiply the multipliers:

$1.04 \times 0.97 = 1.0088 \rightarrow$ **+0.88%**.

IT LOOKS LIKE

+10% then - 10%

-20% then - 30%



$1.04 \times 0.97 = 1.0088$: up 0.88%

How do I do it?

1 Each multiplier

+4% $\rightarrow$ 1.04, -3% $\rightarrow$ 0.97.

2 Multiply them

One overall multiplier.

3 Read it

$1.0088 \rightarrow$ +0.88%; $0.56 \rightarrow$ -44%.

4 Decide

Answer the claim with the number.

MULTIPLY THE MULTIPLIERS Percentages don't add.

V5 Repeated & Combined Changes

VARIANT 5 · SEE IT DONE, THEN TRY


 Your notes
WHEN YOU SEE → YOU DO

"increases by 4% then decreases by 3%" → 1.04×0.97

"Is this the same as a 50% reduction?" → **Overall multiplier, then compare**

"Find the overall percentage change" → **(multiplier - 1) × 100**

WORKED EXAMPLE **PRACTICE**

Percentages · Hard Q18 · 3 marks

A price increases by 4% and then decreases by 3%. Ali says the overall change is a 1% increase. Is he right?

STEP 1 1.04, 0.97 ◀ each multiplier

STEP 2 $1.04 \times 0.97 = 1.0088$ ◀ multiply them

STEP 3 $1.0088 \rightarrow +0.88\%$ ◀ read it

ANSWER No — the increase is 0.88%, not 1% ✓ try £100: £104, then £100.88

WATCH OUT

✗ $+4\% - 3\% = +1\%$

✓ $1.04 \times 0.97 = 1.0088$

The 3% is taken from the bigger, new amount.

EXAM TIP

Test with £100 — it makes the overall change easy to see.

Check yourself

calculator first · answers on page 17

5.1

Dr Eslam

A price rises by 10% then falls by 10%. Overall percentage change?

Answer:

5.2

Practice · Medium

A price is cut by 20%, then by 30%. Is that a 50% reduction?

Answer:

5.3

Practice · Hard

A value falls by 4%, then by 2.5%. Overall percentage decrease?

Answer:

V6

VARIANT 6

Percentages with Fractions & Ratio

Split the total first (ratio or fractions), then take percentages of each part.

SEEN IN PAST PAPERS

8 / 41

PRACTICE QUESTIONS: 7

 Your notes

Came up in: Oct/Nov 2020 P1HR Q10 • Jan 2021 P2H Q10 • May 2021 P2H Q6 • Jun 2022 P1HR Q4 • Jun 2022 P1H Q13 • Jun 2022 P2H Q8 • May 2024 P1H Q14 • Jun 2024 P2H Q8

? What is it?

In plain words

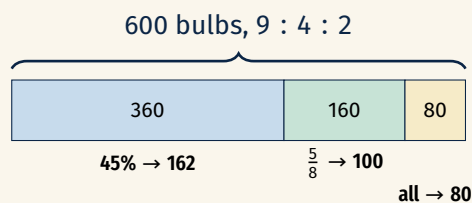
Several steps mixed together.

First **split the total** (ratio parts or fractions). Then take the **percentage or fraction of each part**. Then add what is asked.

IT LOOKS LIKE

9 : 4 : 2

45%

 $\frac{5}{8}$ $\frac{3}{8}$ of the total

How do I do it?

1 Split the total

Ratio: total ÷ parts. Fractions/%: of the whole.

2 Each part

Apply its percentage or fraction.

3 Combine

Add the parts the question wants.

4 Form

Number, fraction (simplest) or percentage — as asked.

SPLIT FIRST, THEN % Each percentage is of its own group.

V6 Percentages with Fractions & Ratio

VARIANT 6 • SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"tulip : crocus : daffodil = 9 : 4 : 2" → **Split 600 into 360, 160, 80**

"45% of the tulip bulbs ..." → **45% of 360 — not of 600**

"Give your answer as a fraction in its simplest form" → **Part ÷ total, simplify**

WORKED EXAMPLE **PAST PAPER**

June 2024 • Paper 2H • Q8 • 5 marks

600 bulbs: tulip : crocus : daffodil = 9 : 4 : 2. 45% of tulips, $\frac{5}{8}$ of crocuses and all daffodils are yellow. How many bulbs are yellow?

STEP 1 $600 \div 15 = 40$: 360, 160, 80 ◀ split the total

STEP 2 $0.45 \times 360 = 162$, $\frac{5}{8} \times 160 = 100$ ◀ each part

STEP 3 $162 + 100 + 80 = 342$ ◀ combine

ANSWER 342 yellow bulbs

✓ all counts are whole numbers ✓

WATCH OUT✗ 0.45×600 ✓ 0.45×360

Each percentage is of its own group, not of all 600.

EXAM TIP

8 of the 41 past-paper parts, 4–5 marks.
Write the split line first — it is the first method mark.

Check yourself

calculator first • answers on page 17

6.1 Jun 2022 P1HR

180 students: 15 French, 45% German, the rest Italian or Spanish. Italian/Spanish as a % of French/German?

Answer:

6.2 Jun 2022 P1H

Payment 1 is $\frac{3}{8}$ of the total, payment 2 is 45%, payment 3 is \$406. Find payment 2.

Answer:

6.3 Jan 2021 P2H

Teams A : B = 3 : 4. $\frac{4}{5}$ of A and 24% of B work full time. What fraction work full time?

Answer:

CHECK YOURSELF

Answers



Your notes

Every answer below was checked with sympy (check_answers.py).

1.1 £102**1.2** \$484**1.3** £276**2.1** 22.5%**2.2** 11.8%**2.3** $P = 30$ **3.1** 12.5%**3.2** 44%**3.3** 15%**4.1** £16.25**4.2** 275 euros**4.3** 7500 rupees**5.1** 1% decrease**5.2** No: a 44%
reduction**5.3** 6.4% decrease**6.1** 87.5%**6.2** \$1044**6.3** $\frac{12}{25}$

THE WHOLE TOPIC ON ONE PAGE

Percentages cheat sheet



Your notes
V1 Increase/decrease 4/41

$$550 \times 0.88 = 484$$

Original $\times$ multiplier.**V2 % change 10/41**

$$\frac{3462}{228\,314} \times 100$$

Change $\div$ original.**V3 % profit 3/41**

$$\frac{1020}{1980} \times 100$$

Build cost and income; $\div$ cost.**V4 Reverse 16/41**

$$20.40 \div 0.85 = 24$$

After the change $\rightarrow$ divide.**V5 Repeated**

$$1.04 \times 0.97 = 1.0088$$

Multiply the multipliers.

V6 With ratio 8/41

$$360, 160, 80$$

Split first, then % of each part.

**TOP 3 LOST MARKS**

1. reverse: $\times 1.15$ instead of $\div 0.85 \times$
2. dividing by the new value $\times$
3. adding percentages $\times$

BOTTOM LINE Find the multiplier $\rightarrow$ forwards: $\times$ • backwards: $\div$ • % change: $\div$ original • check forwards.